

When Personalization Fails: Testing Whether LLM Personas Actually Predict Human Decisions

A pre-registered, sealed evaluation of persona fine-tuning against a strong ideological baseline in U.S. Supreme Court voting

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PRODUCT C — computing-oriented re-framing. This is the computing draft derived from the parent manuscript *Legally Subjective: How Much Vote Predictability Survives a Strong Ideological Baseline?* (23 pages, law-review framing, revision 3, October 2026). The law is the application domain here, not the subject. Start any actual approach to CACM from CACM_PITCH.pdf; contributed pieces are proposed to the editors first, and this draft is the text the editors would be invited to see.

1. The premise, and the evaluation gap

Personalization is one of the most commercially consequential promises of large language models: fine-tune the model on the writings of a specific person, the story goes, and it will not merely sound like that person but anticipate them — their choices, their votes, their decisions. The claim is rarely tested where it can actually fail cleanly. Published demonstrations of persona-style agents seldom face a strong non-LLM baseline, seldom control for training-data leakage, and almost never commit their predictions before the ground truth exists. The result is an evaluation gap: the community knows how to make models imitate people, but not whether imitation adds predictive information.

We built the experiment that closes this gap for one high-stakes, well-instrumented domain: voting by U.S. Supreme Court justices, where every vote of every justice is published and coded, and where a simple ideological rule is famously hard to beat. The question is narrow and falsifiable: does a persona — a biographical prompt plus a QLoRA adapter fine-tuned on a justice’s own past opinions — improve out-of-sample prediction of that justice’s votes, relative to a zero-shot prompt and to an ideological prior? In this sealed test, the answer is no: personalization showed no detectable improvement on the primary endpoint, while measurably changing the model in ways that matter for anyone deploying personalized LLMs.

2. An experimental substrate built for evaluation

The substrate is a frozen, hash-chained corpus assembled from two public sources: the Supreme Court Database (vote coding, case metadata) and CourtListener (full opinion texts). It covers 569 argued cases from OT2015 through OT2023, thirteen justices, and 4,569 individually coded votes. The opinion text layer was normalized and deduplicated from 1,778 collected documents to 793 distinct texts with full provenance; the SHA-256 of every raw source file is recorded at download and recomputed by the corpus build, so provenance is structural rather than narrative. The corpus, its build scripts, and its exclusions (ghost docket groups, consolidated matters

without SCDB matches) are all public and documented in the repository.

3. The baseline that sets the bar

Every predictive claim in this study is measured against pre-registered baselines — the “price of admission” that any challenger must pay before personalization is even discussed. The strongest is a per-justice ideological rule (condition D’s parent, baseline B4): map each justice to a fixed liberal/conservative direction and predict accordingly. On the held-out window it reaches 63.66% of individual votes (bootstrap interval [0.6132, 0.6593]). Three families of structured-feature challengers trained on the same corpus — including grouped cross-validated classifiers over case metadata — fail to clear that bar. That failure is itself the methodological headline: before any language model enters the room, a two-parameter ideological rule already captures most of what is predictable.

4. Four conditions, one sealed exam

The written-record test is a sealed examination on fifty hash-frozen 5–4 decisions — the Court’s hardest cases, where the attitudinal signal is deliberately weakest. Four conditions predicted the sealed cases blind: **A**, zero-shot prompting of the base model as a neutral legal analyst; **B**, the persona — the same base model with a justice-specific biographical prompt and a QLoRA adapter fine-tuned on that justice’s past opinions; **C**, retrieval of the justice’s relevant past writings into the prompt; and **D**, the ideological prior re-fit on the transparent window. Predictions were generated, frozen, and hash-locked before any answer key existed, and scored exactly once. Comparisons on the paired sealed set use exact McNemar tests.

The seal is enforced by infrastructure, not by promises: results/m4_exam.lock records both the repository head and the SHA-256 of the frozen predictions; the sealed fifty were drawn by a deterministic seeded procedure whose seed derivation and list hash are committed; and regenerating the sealed list after training would be visible in the git history alone. Because a sealed exam that can be re-taken is not sealed, the GPU-dependent stages are frozen as executed artifacts rather than re-runnable scripts.

5. Leakage discipline

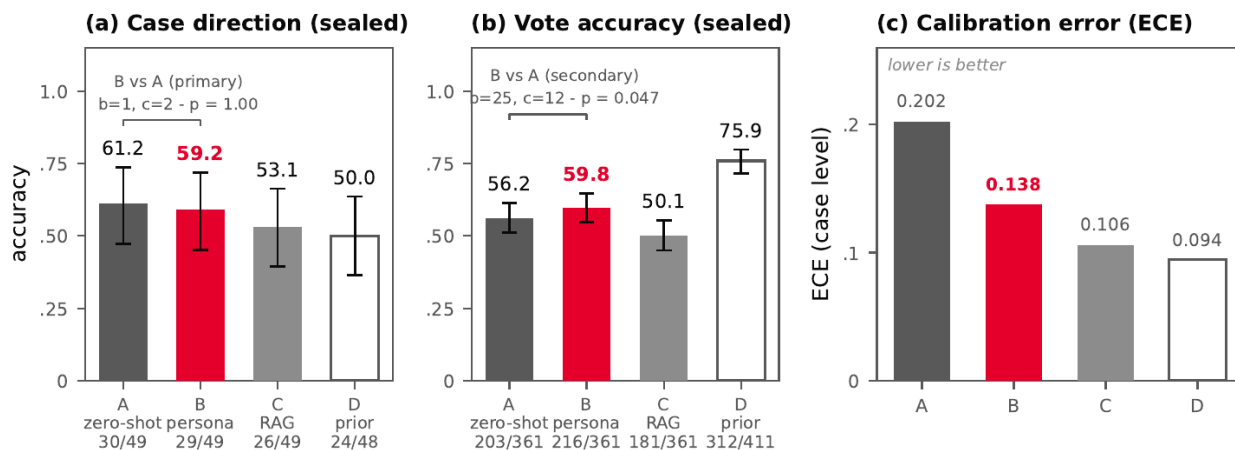
Leakage is the silent killer of personalization studies: the model may have seen the test cases in pretraining, and the persona may have absorbed the very texts being predicted. We control both. The persona training pool was audited row by row: 477 training rows, zero sealed rows, zero out-of-bounds rows. The fourteen-check zero-leak audit is re-executed by continuous integration on every push to the public repository. Contamination of the base model by pretraining exposure cannot be excluded — thirty of the fifty sealed decisions predate the base model’s knowledge cutoff — but it is common to every language condition and therefore cancels in the paired comparison, which is the design’s whole point.

6. What the sealed exam said

Condition	Case-direction accuracy	Vote-level accuracy
A — zero-shot	61.22%	56.23%
B — persona (prompt + QLoRA adapter)	59.18%	59.83%

Condition	Case-direction accuracy	Vote-level accuracy
C — retrieval	53.06%	50.14%
D — ideological prior	50.0%	75.91%

Sealed exam, $n = 50$ hash-frozen 5–4 decisions (49 paired for McNemar). Case-level: majority-of-votes direction. Vote-level: individual justice votes.

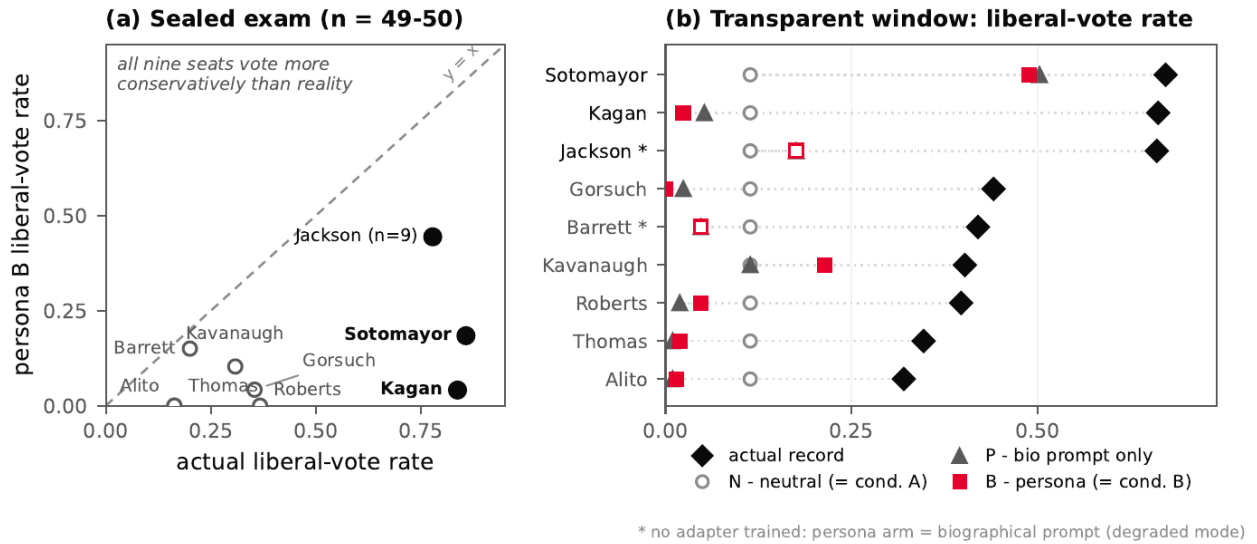


Primary sealed-exam results (parent manuscript, fig. 5): case-direction and vote-level accuracy by condition, with confidence machinery. The persona never separates from the zero-shot on the primary endpoint.

Personalization does not detectably improve prediction. On the primary endpoint — case-direction accuracy — the persona scores 59.18% against the zero-shot’s 61.22%; the exact McNemar test on the paired decisions diverges on only three cases (one in the persona’s favor, two against; $p = 1.00$). Meanwhile the ideological prior collapses to an exact coin flip at case level (50.0%) on these one-vote-margin decisions while still dominating every language condition at vote level (75.91% against 50–60%). Case-direction kappa against the always-conservative majority class is 0.09 for A, 0.02 for B, 0.06 for C: on 5–4 cases, every condition — language model and statistical rule alike — is a small perturbation of “always guess conservative.” The same machinery scored on the transparent window yields 55.8% at case level, against 50.0% on the sealed set: the hardest cases really are harder, and nothing in the language stack bridges the gap.

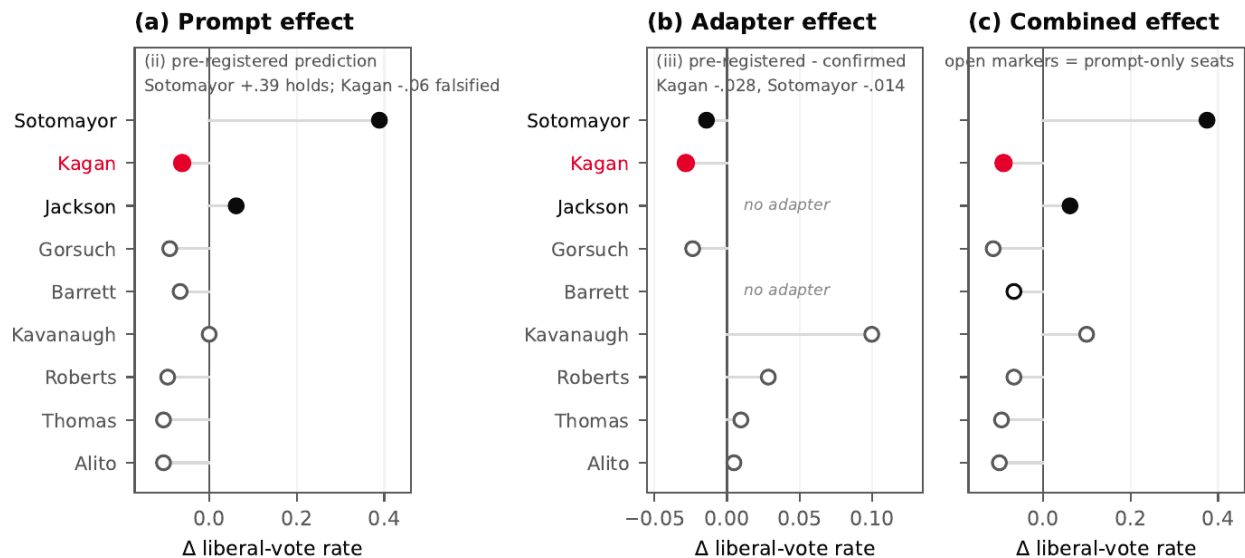
7. What personalization did change

A negative endpoint with a positive mechanism is worth more than a bare negative, and the persona does change the model — systematically. Measure each seat’s simulated liberal-vote rate (a dial that needs no ground truth): every persona, including those imitating liberal justices, votes more conservatively than the justice it imitates. The persona-predicted liberal rate never exceeds 0.44 while the actual rates span 0.16–0.86; Kagan’s persona votes liberal on 4.1% of her sealed cases where the justice herself votes liberal on 83.7%. The mechanism is compression: the persona’s mean predicted probability of a liberal vote spans only 0.16–0.39 across seats, re-anchoring the Court’s real ideological span into a narrow band whose upper edge stays below one half.



The ideological dial (parent manuscript, fig. 6): actual vs. simulated liberal-vote rates per seat. Every persona sits strictly below the diagonal; the decomposition panel isolates the neutral arm (N), the biographical prompt (P), and the full persona (B).

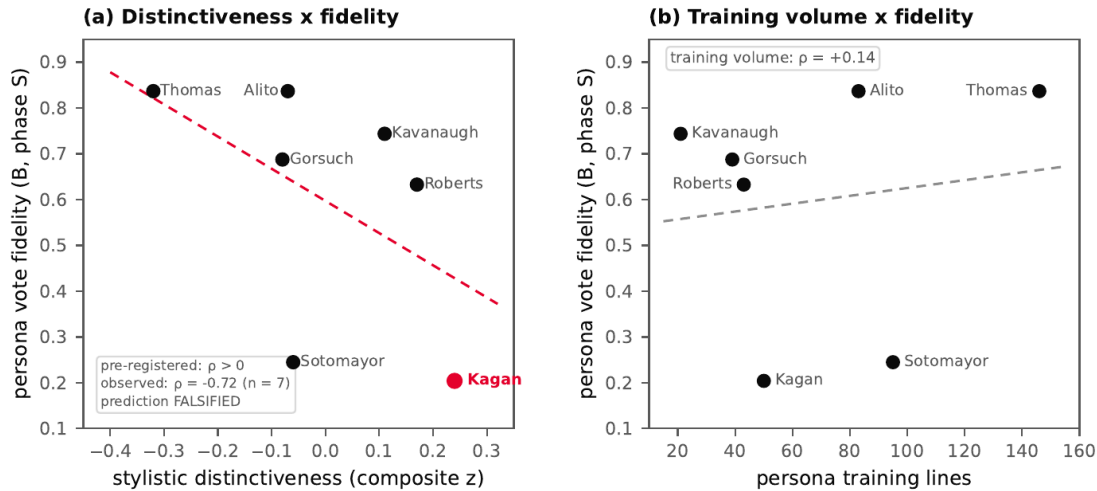
A causal decomposition experiment (E1, pre-registered, transparent window, $n = 211$ casefiles, seed 271828) splits the persona into its two components. The neutral arm has a strong conservative default (0.114 liberal calls for every seat). The biographical prompt does most of the moving — lifting Sotomayor to 0.50 (a +0.39 prompt effect) while paradoxically failing for Kagan (−0.06, falsifying the second pre-registered prediction). The adapter’s marginal effect on top of the prompt never exceeds 0.0996 of the dial, and is negative for both pre-registered liberal targets: the adapter actually erases part of the prompt’s gain. Identity injected in the prompt is not behavior learned by the weights.



E1 decomposition (parent manuscript, fig. 7): prompt effect $\Delta(P-N)$, adapter effect $\Delta(B-P)$, and combined $\Delta(B-N)$ per seat. The adapter’s marginal effect is small everywhere (≤ 0.10) and negative for the pre-registered targets.

Two further pre-specified probes close the diagnosis. An auto-coherence probe (E3) shows the persona misvoting on the very training cases whose texts it absorbed — imitation of prose does not even reproduce the

votes that produced the prose. And the natural stylistic explanation — that personas work better for justices with more distinctive writing styles — was pre-registered as a fidelity account and then falsified: the Spearman correlation between stylistic distinctiveness and persona quality is $\rho = -0.72$, the wrong sign. What remains is compression: personalization moves individual votes (nominally $p = .047$, unadjusted) without improving case-direction prediction.



E2 (parent manuscript, fig. 8): stylistic distinctiveness does not predict persona quality; the pre-registered stylistic account is falsified ($\rho = -0.72$).

8. Reproducibility as a first-class result

Every CPU-side number in this study regenerates from the public repository with three commands and no credentials: the corpus build (raw sources to frozen corpus with recomputed hashes), the baselines, and the challenger suite with its grouped cross-validation and McNemar tests — all seeded, with seeds printed in the outputs. The GPU-dependent stages are frozen as executed artifacts, and the freeze is the point: the sealed predictions carry their own SHA-256 inside a lock that records the repository head, so any post-hoc tampering is detectable from the committed history alone. The fourteen-check zero-leak audit re-runs in continuous integration on every push. A reviewer can walk the entire chain — paper, code, corpus, frozen artifacts, predictions, regeneration — in minutes.

9. Implications for evaluation practice

- 1. Put the baseline first.** A two-parameter ideological rule predicted 63.7% of votes; sophisticated challengers failed to clear it. Any personalization claim that has not paid this price is not a prediction result, it is a demo.
- 2. Seal the test.** Predictions frozen and hash-locked before the answer key exists, scored exactly once — this is cheap to do with git and SHA-256, and it converts an evaluation from narrative to evidence.
- 3. Audit for leakage, mechanically.** Row-level audits of the training pool (477 rows, zero sealed, zero out-of-bounds) re-run in CI; leakage controls belong in infrastructure, not in the limitations section.
- 4. Decompose personalization.** Prompt identity and fine-tuned weights are separable causes; in this study the prompt did most of the moving and the adapter’s marginal effect never exceeded 0.0996. Claims about

“fine-tuned personas” that have not been decomposed are claims about a mixture.

5. Report the negative. A clean null from a sealed, reproducible experiment is a real contribution — especially when the same experiment shows what the intervention does change (here: a systematic conservative shift and probability compression).

10. Why Supreme Court votes

The domain is chosen for instrumentation, not for provocation. U.S. Supreme Court voting offers complete, public, per-vote ground truth over decades — a rarity for human decisions that matter — plus a famous baseline result and a live debate about machine prediction of judicial behavior. The subjectivity of judges is, in theory, the last thing that should automatize cleanly; measuring whether it does is a well-posed question with reusable lessons for any domain where LLMs are personalized to imitate human decision-makers.

Availability and declarations

Corpus, scripts, adapters, frozen artifacts, sealed predictions, and audit reports are public at github.com/Vitalcheffe/legally-subjective. The research used large-language-model coding and writing assistants under the authors’ direction; every reported number regenerates from committed scripts over frozen artifacts, and the disclosure appears in the parent manuscript’s declarations. Funding: none — the project ran on free-tier services and public data. The authors are high-school students who conducted the research independently; their school is named for identification only and does not endorse the research.

SOURCE: Parent manuscript: Legally Subjective: How Much Vote Predictability Survives a Strong Ideological Baseline? — working paper, revision 3, October 2026 (23 pages), same repository. Figures 5–8 are reused from the parent manuscript without modification.